

EXAM – SOLUTION

This is a 3 hours exam. Class slides and any handwritten material are allowed

I – TRUE, FALSE, UNCERTAIN? (1/3 of the points)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 lines per question.

Note: There is often some arbitrariness in choosing “True” or “False” rather than “Uncertain”, as it depends on the level of generality one chooses. Therefore, what really matters for evaluation is the full answer.

1 – When studying the business cycle, the Hodrick-Prescott better fits the data.

FALSE. The Business Cycle extraction is not a matter of estimation, but of *representation*. There is an infinite number of trend/cycle decomposition of a non stationary time series. They all fit the data the same way (meaning that trend + cycle is always equal to the original series). The choice of the relevant filter is an economic one. It happens that the HP filter with smoothing parameter $\lambda = 1600$ does remove low frequency cycles (more than 10 years), which is about the average length of the NBER business cycles.

2 – In a growth model, it is optimal to be along a balanced growth path.

FALSE. The BGP is a path that is *convenient to study*, because growth rates are constant. It makes sense to study *the* BGP in models where it is unique and in which other paths converge to the BGP. In some other growth models, it is an arbitrary restriction to study BGP (for example when non BGP solution exists).

3 – (*This question counts the double of any other*) In the Ramsey model, starting from a steady state, an increase in the depreciation rate decreases consumption in the short run and increases it in the long run (use a phase diagram and draw the path of c and k .)

FALSE. The phase diagram is below. The differential equations that govern the dynamics of the Ramsey model are

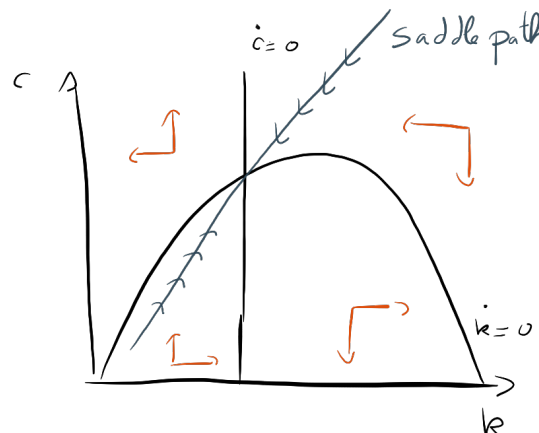
$$\frac{\dot{c}_t}{c_t} = \sigma(f'(\frac{k_t}{L}) - \delta - \rho) - g.$$

and

$$\dot{k}_t = \bar{L}f(\frac{k_t}{L}) - (\delta + g)k_t - c_t$$

Figure 1 is showing the phase diagram with the saddle path. Figure 2 shows how the phase diagram is modified when

Figure 1: Ramsey with depreciation shock - Phase diagram



δ increases. An increase in δ decreases k in the long run, so that consumption is necessarily lower at the new steady state compared to the old one. In the short run, depending on the slope of the saddle path (see case 1 and case 2 on Figure 3), consumption can jump up or down. This depends of the relative strength of wealth and substitution effect. We can go further (of course not during the exam!) by looking at a quantitative example. Assume time is discrete,

Figure 2: Ramsey with depreciation shock - Phase diagram changes

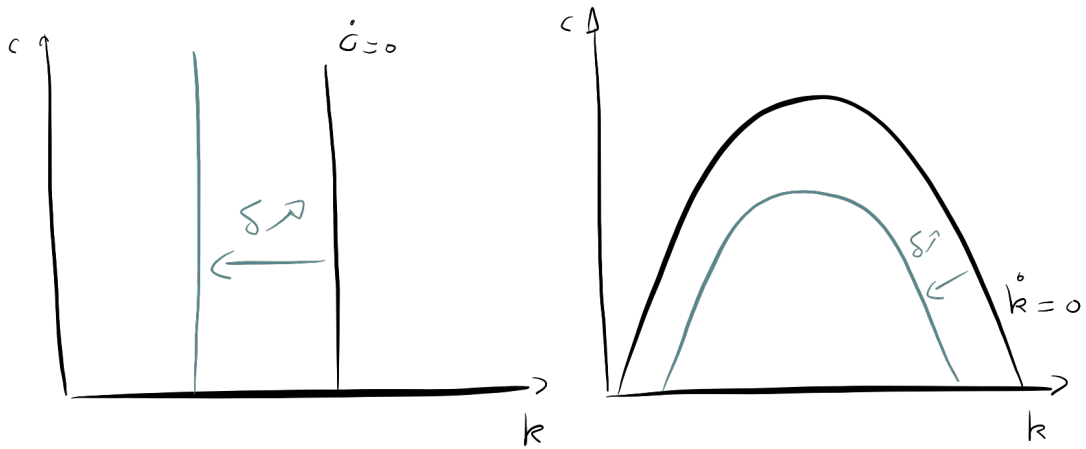
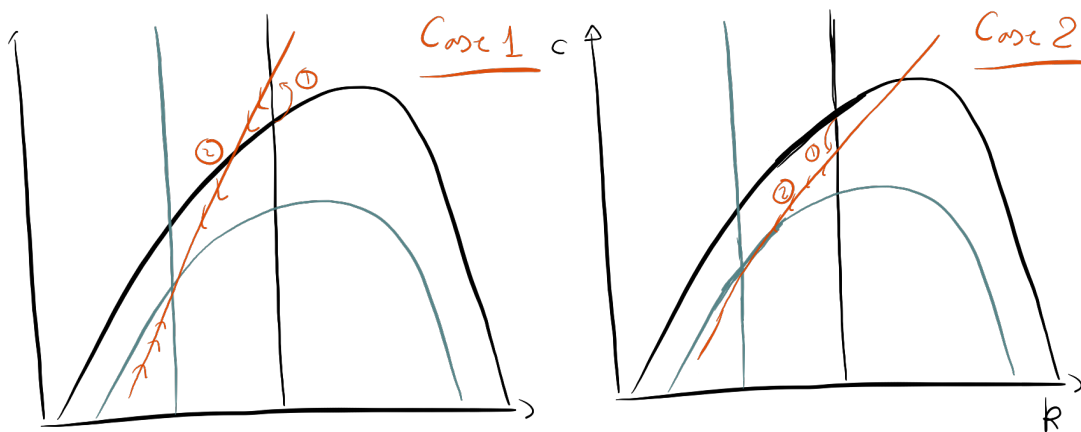
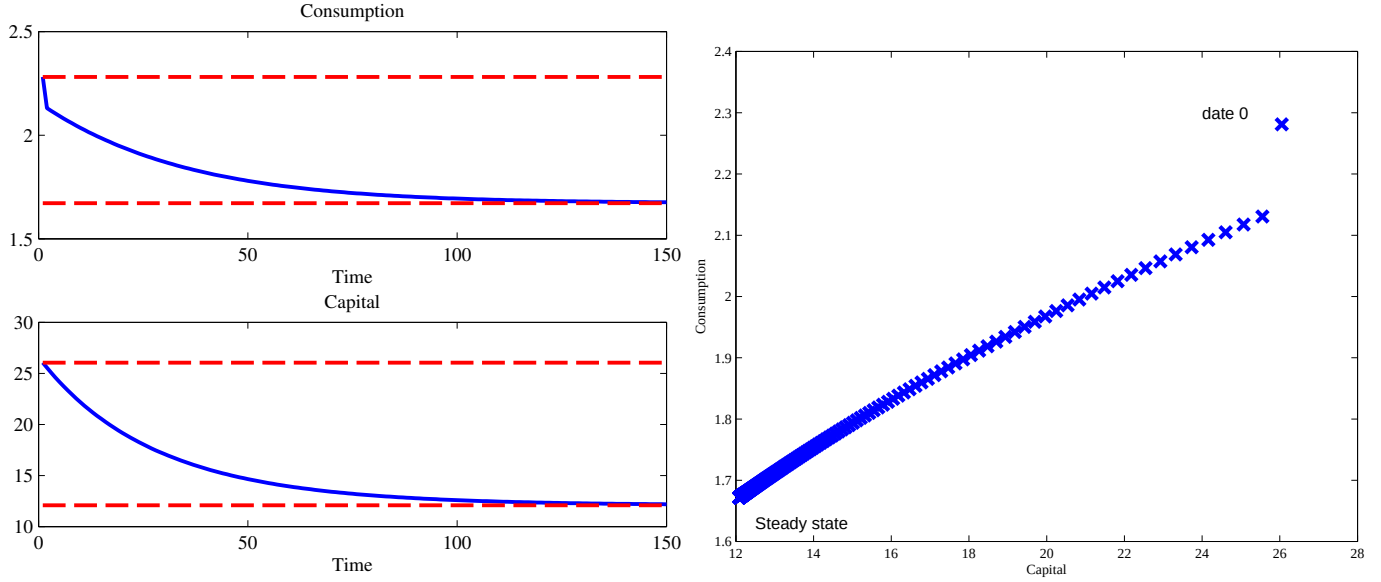


Figure 3: Ramsey with depreciation shock - Possible paths



$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ with $\sigma = .4$ (high intertemporal substitution) or $\sigma = 4$ (low intertemporal substitution), $f(k) = k^\alpha$ with $\alpha = .33$. Finally, depreciation δ is .025 at the initial steady state, and is increased unexpectedly and permanently in period 1 to the value .05.

Figure 4: Ramsey with depreciation shock - The case $\sigma = 4$ (low intertemporal substitution) (Note: On the right panel, each $\times$ marker represents couple (k_t, c_t) for a given period t .)



4 – Search frictions cause unemployment to be too high.

TRUE/FALSE. Of course, compared to a frictionless benchmark in which unemployment is zero, this statement is true. Once we take as given search frictions, unemployment becomes an input in the matching function, *i.e.* the creation of new jobs. Thus there is no presumption that the optimal unemployment level is zero. To have a low unemployment rate, vacancies must be high. But vacancies consume resources, so that potentially we can have too many vacancies and to low unemployment. On the other hand, the unemployed are also costly because they do not produce. It is also possible to have too much unemployment. There are two central market failures in the matching model : congestion externalities and appropriability problems.

The congestion externalities are as follows. An increase in search exerts positive externalities on the other side of the market. Unemployment reduces the length of vacancies ; conversely vacancies reduce the length of unemployment. Because the matching function is concave, search activity exerts a negative externality on the same side of the market : more unemployment reduces the job finding rate.

The appropriability problem can be summarized as follows: a surplus arises from turnover costs, and part of it surplus is appropriated by the worker in the bargaining process. The firm only appropriates a fraction of the surplus created by a vacancy, but pays the entire vacancy cost. Hence the incentives to post vacancies are too low. This is what we call a hold-up problem. A symmetrical issue arises on the worker's side when evaluating the value of being unemployed : only a fraction of the value of future jobs will be appropriated by the worker, hence the value of looking for a job (which is equal to the opportunity cost of working now) is underestimated.

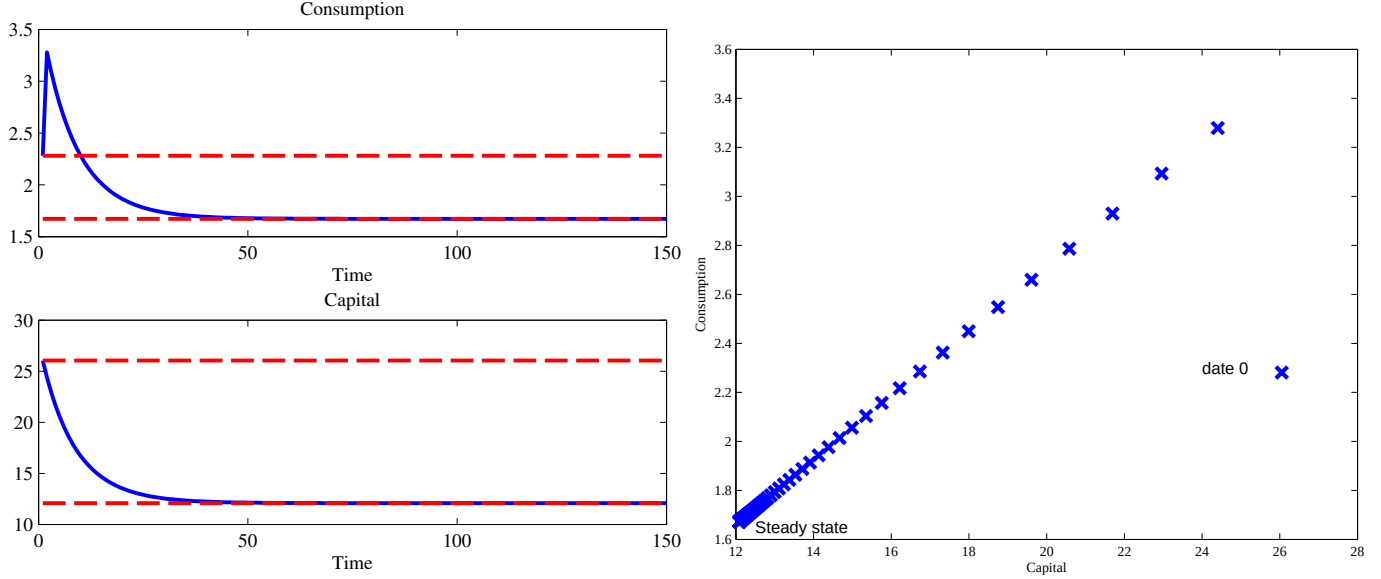
5 – Sustained growth is obtained when production functions have constant returns.

FALSE. Asymptotically, we must have returns to accumulated factors equal to or greater than one for at least one accumulable factor of production.

6 – In endogenous growth models, better enforcement of intellectual property rights unambiguously raises welfare.

FALSE. One the one hand, in R&D models, innovators typically do not entirely appropriate the social surplus from their innovation. They get profits but part of the surplus goes to the consumers. This tends to make innovation too low compared with the optimum. Better enforcement of intellectual property rights foster R&D, therefore growth, and is a good thing for welfare. But on the other hand, ideas are non rival goods. The monopoly distortion in price-setting that is created by intellectual property rights tends to push prices above marginal cost. In equilibrium this means that real wages are too low relative to the social optimum, which is bad for welfare.

Figure 5: Ramsey with depreciation shock - The case $\sigma = .4$ (high intertemporal substitution)(Note: On the right panel, each $\times$ marker represents couple (k_t, c_t) for a given period t .)



7 – Durable goods consumption is a smooth variable in the business cycle.

Uncertain. The consumption of durable goods *services* is smooth if preferences are convex. But what is measured in the data is not consumption of durable goods services but *expenditures* in durable goods. Those expenditures are lumpy. For example, one replaces her car only (say) every five years, not continuously. This lumpiness is largely smoothed out by aggregation. But the timing of expenditures in durable goods is also engogenous, and sensitive to business cycle conditions. in a recession, everybody tend to delay the replacement of a car. Therefore, the durable consumption series in national accounts looks like investment, and is more volatile than output.

8 – Calibration of an RBC model is not so far from estimation.

UNCERTAIN. On the one hand, calibration is about fitting some moments by choosing accordingly some parameters. It is about minimizing a distance, and therefore a kind of estimation. On the other hand, there is often as many parameters as moments to match, so that no inference or statistical test can be done (the “R²” is one. On top of that, calibration is done on a set of moments different from the ones the modeler is interested with. For example, RBC models are calibrated using steady star moments, and are used to produce business cycles moments.

9 – Consider the following stochastic difference equation: $x_t = \beta E_t x_{t+1} + y_t$, where y_t is an exogenous stationary stochastic process. This equation has a unique stationary solution only if $|\beta| > 1$.

FALSE. The condition for uniqueness is $|\beta| < 1$. Take the model $x_t = \beta E_t x_{t+1} + y_t$. This implies $x_{t+i} = \beta E_t x_{t+i+1} + y_{t+i} \quad \forall i$. Taking expectations

$$E_t x_{t+i} = \beta E_t x_{t+i+1} + E_t y_{t+i}$$

Substitute forward (*forward integration*):

$$x_t = \beta^k E_t x_{t+k} + \sum_{j=0}^{k-1} \beta^j E_t y_{t+j}$$

. Impose that $E_t x_{t+k}$ is bounded $\forall k$. Then if $|\beta| < 1$, then $\lim_{k \rightarrow \infty} \beta^k E_t x_{t+k} = 0$, hence

$$x_t = \sum_{j=0}^{\infty} \beta^j E_t y_{t+j}.$$

This is the unique stationary solution to the rational expectations model.

If $|\beta| > 1$, then the solution cannot be unique. For any solution x_t , $x_t + b_t$ is also a solution under some conditions

on b . Those conditions are that b_t is a random variable (sunspot) satisfying

$$b_{t+1} = \frac{1}{\beta}b_t + u_{t+1},$$

with u any shock such that $E_t u_{t+1} = 0$. Furthermore, we can construct at least one non explosive solution by picking $x_{t+1} = \frac{1}{\beta}(x_t - \eta_t)$. Hence in this case we have a continuum of equilibria.

II – GROWTH MODEL WITH HUMAN CAPITAL (*1/3 of the points*)

Consider an economy with a single infinitely lived household, whose preferences are

$$\int_0^\infty e^{-\rho t} \frac{c_t^{1-\sigma}}{1-\sigma} dt$$

There are two sectors in the economy. Sector 1 produces consumption and capital goods:

$$G(k_{1t}, h_{1t}) = c_t + I_{1t}$$

where $I_{1t} = \dot{k}_t + \delta k_t$ and $k_t = k_{1t} + k_{2t}$. Sector 2 produces human capital:

$$H(k_{2t}, h_{2t}) = I_{2t}$$

where $I_{2t} = \dot{h}_t + \delta h_t$ and $h_t = h_{1t} + h_{2t}$. It is assumed that G and H are constant returns to scale. h_0 and k_0 are given.

There is a government that taxes capital and human capital income. It rebates revenues as a lump-sum transfer T_t . The budget constraint is given by

$$T_t = \tau_k q_{kt} k_t + \tau_h q_{ht} h_t.$$

There is a representative firm in each sector. Firms rent capital and human capital at rental prices q_{kt} and q_{ht} , respectively. Good 1 is the numéraire. The price of good 2 is p_t .

The household's budget constraint is given by:

$$c_t + I_{1t} + p_t I_{2t} = (1 - \tau_k) q_{kt} k_t + (1 - \tau_h) q_{ht} h_t + T_t$$

1 – Derive firms and households first order conditions.

For the firm of sector 1, $\Pi_{1t} = G(k_{1t}, h_{1t}) - q_{kt} k_{1t} - q_{ht} h_{1t}$ and FOCs are

$$\begin{aligned} G_1 &= q_{kt} \\ G_2 &= q_{ht}. \end{aligned}$$

For the firm of sector 2, $\Pi_{2t} = p_t H(k_{2t}, h_{2t}) - q_{kt} k_{2t} - q_{ht} h_{2t}$ and FOC are

$$\begin{aligned} p_t H_1 &= q_{kt} \\ p_t H_2 &= q_{ht} \end{aligned}$$

For the household, the hamiltonian writes

$$\mathcal{H}_t = \frac{1}{1-\sigma} ((1 - \tau_k) q_{kt} k_t + (1 - \tau_h) q_{ht} h_t + T_t - I_{1t} - p_t I_{2t})^{1-\sigma} + \lambda_t (I_{1t} - \delta k_t) + \mu_t (I_{2t} - \delta h_t)$$

The state variables of that problem are k and h , the controls I_1 and I_2 and the co-state λ and μ . FOCs are:

$$u'(c_t) = \lambda_t \tag{1}$$

$$p_t u'(c_t) = \mu_t \tag{2}$$

$$-\dot{\lambda}_t + \rho \lambda_t = u'(c_t)(1 - \tau_k) q_{kt} - \lambda_t \delta \tag{3}$$

$$-\dot{\mu}_t + \rho \mu_t = u'(c_t)(1 - \tau_h) q_{ht} - \mu_t \delta \tag{4}$$

2 – Derive the household's Euler equation, that can be written as

$$\frac{\dot{c}_t}{c_t} = \frac{r_t - \rho}{\sigma}$$

where r can be expressed in two different ways, depending on whether capital or human capital is considered.

From (1), one gets $c_t^{-\sigma} = \lambda_t$. Fully differentiating with respect to time, one obtains

$$\frac{\dot{\lambda}_t}{\lambda_t} = -\sigma \frac{d \log c_t}{c_t} \quad (5)$$

Similarly, equation (2) implies

$$\frac{\dot{\mu}_t}{\mu_t} - \frac{\dot{p}_t}{p_t} = -\sigma \frac{d \log c_t}{c_t} \quad (6)$$

Now take (3), divide by λ , use (1) and (5) to obtain

$$\frac{\dot{c}_t}{c_t} = \frac{((1 - \tau_k)q_{kt} - \delta) - \rho}{\sigma}$$

Similarly, (4), (2) and (6) imply

$$\frac{\dot{c}_t}{c_t} = \frac{\frac{\dot{p}_t}{p_t} + \frac{1}{p_t}((1 - \tau_h)q_{ht} - \delta) - \rho}{\sigma}$$

We see from those two equations that one has

$$\frac{\dot{c}_t}{c_t} = \frac{r_t - \rho}{\sigma}$$

with the two expressions for r

$$\begin{aligned} r_t &= ((1 - \tau_k)q_{kt} - \delta) \\ &= \frac{\dot{p}_t}{p_t} + \frac{1}{p_t}((1 - \tau_h)q_{ht} - \delta) \end{aligned}$$

3 – Derive four equations that solve for the balanced growth values of g , z_1 , z_2 , r where g is the balanced growth rate of (c, k, h) and $z_i = k_i/h_i$. Note that p is constant on the balanced growth path. Remember that, with constant returns to scale, marginal products are functions of z_i .

Along a balanced growth path with real growth rate g , we have

$$\begin{aligned} g &= \frac{r - \rho}{\sigma} \\ r &= (1 - \tau_k)G_1(z_1, 1) - \delta \\ r &= (1 - \tau_h)H_2(z_2, 1) - \delta \\ \frac{G_1(z_1, 1)}{G_2(z_1, 1)} &= \frac{H_1(z_2, 1)}{H_2(z_2, 1)} \end{aligned}$$

where we use $\dot{p}/p = 0$ along a BGP.

4 – For the special case where $H(k_{2t}, h_{2t}) = Bh_{2t}$, show that taxes on sector 1 do not affect the balanced growth rate. What is the intuition for this result?

With $H(k_{2t}, h_{2t}) = Bh_{2t}$, $H_2 = B$. Therefore, $r = (1 - \tau_h)B - \delta$ and $g = \frac{r - \rho}{\sigma} = \frac{(1 - \tau_h)B - \delta - \rho}{\sigma}$. We see that the interest rate and the growth rate are fully determined by looking at sector 2, because of the linearity of H . Taxing sector 1 will not affect the growth rate, but the levels, as a change in τ_k will cause an adjustment in z_1 such that $r = (1 - \tau_k)G_1(z_1, 1) - \delta$ still holds.

III – MORTENSEN-PISSARIDES (1/3 of the points)

Consider the following version of the matching function giving the number of matches M :

$$M = aV^\alpha U^{1-\alpha} \quad (1)$$

where V is the number of vacancies and U is the number of unemployed job-seekers. Assume that $\alpha = 1/2$.

1 – Define $\theta = \frac{V}{U}$ as the measure of labor market tightness. By using the matching function (1), write the rate at which a firm fills a vacant job $\frac{M(V, U)}{V}$ and the rate at which an unemployed worker finds a job $\frac{M(V, U)}{U}$ as a function of θ . Denote the former with $q(\theta)$ and the latter with $p(\theta)$.

$$q(\theta) = \frac{M(V, U)}{V} = M\left(1, \frac{U}{V}\right) = M\left(1, \frac{1}{\theta}\right) = a\theta^{-\frac{1}{2}}$$

and

$$p(\theta) = \frac{M(V, U)}{U} = M\left(\frac{V}{U}, 1\right) = M(\theta, 1) = a\theta^{\frac{1}{2}}$$

2 – Compute $q'(\theta)$ and $p'(\theta)$. Interpret economically the signs of these functions.

Derivatives are $q'(\theta) = -\frac{1}{2}a\theta^{-\frac{3}{2}} < 0$ (congestion) and $p'(\theta) = \frac{1}{2}a\theta^{-\frac{1}{2}} > 0$ (thick market effect).

3 – Denote the exogenous job destruction rate with δ , wage with w , non-labor market income with b , and the discount rate with r . Write the discounted expected utility of an employed person V_e and the discounted expected utility of an unemployed person V_u where you should use the exit rate from unemployment derived previously.

$$rV_e = w + \delta(V_u - V_e) + \dot{V}_e \quad (1)$$

and

$$rV_u = b + p(\theta)(V_e - V_u) + \dot{V}_u. \quad (2)$$

Selecting the only non-explosive solution to the first equation implies $\dot{V}_e = 0$.

4 – Denote the output by worker with y and the cost of holding a vacancy with c . Write the discounted expected profits from a filled job Π_e and the discounted expected profits from a vacant job Π_v where you should use the probability of a match occurring derived previously.

$$r\Pi_e = y - w + \delta(\Pi_v - \Pi_e) + \dot{\Pi}_e \quad (3)$$

and

$$r\Pi_v = -c + q(\theta)(\Pi_e - \Pi_v) + \dot{\Pi}_v. \quad (4)$$

5 – Assuming free entry, use the expressions for Π_e and Π_v to derive the firm's labor demand that gives the relationship between θ and the net production $y - w$. This gives the labor demand equation (LD).

Free entry implies $\dot{\Pi}_v = 0$. (4) implies $\Pi_e = \frac{c}{q(\theta)}$. Taking (3), using $\Pi_v = 0$, selecting the only non explosive solution for Π_e gives $\Pi_e = \frac{y-w}{r+\delta}$. Putting those two expressions of Π_e together, we obtain

$$\theta = \left(\frac{a(y-w)}{c(r+\delta)} \right)^2 \quad (LD)$$

6 – Solve the Nash bargaining problem:

$$\max_w \Omega = \frac{1}{2} \log(V_e - V_u) + \frac{1}{2} \log(\Pi_e - \Pi_v)$$

assuming free entry. Write the wage as a function of θ . This gives the wage equation (WE).

The first order condition of the Nash bargaining problem writes:

$$\frac{d\Omega}{dw} = \frac{1}{2} \left[\frac{\frac{d(V_e - V_u)}{dw}}{V_e - V_u} + \frac{\frac{d(\Pi_e - \Pi_v)}{dw}}{\Pi_e - \Pi_v} \right] = 0$$

Subtracting rV_u from both sides of (1), one gets:

$$V_e - V_u = \frac{w - rV_u}{\delta + r}$$

so that $\frac{d(V_e - V_u)}{dw} = \frac{1}{\delta + r}$.

Subtracting $r\Pi_v$ from both sides of (3), one gets:

$$\Pi_e - \Pi_v = \frac{y - w - r\Pi_v}{\delta + r}$$

so that $\frac{d(\Pi_e - \Pi_v)}{dw} = \frac{-1}{\delta + r}$.

Therefore the FOC rewrites:

$$\frac{1}{\delta + r} \frac{1}{V_e - V_u} - \frac{1}{\delta + r} \frac{1}{\Pi_e - \Pi_v}$$

so that, using $\Pi_v = 0$,

$$\Pi_e = V_e - V_u$$

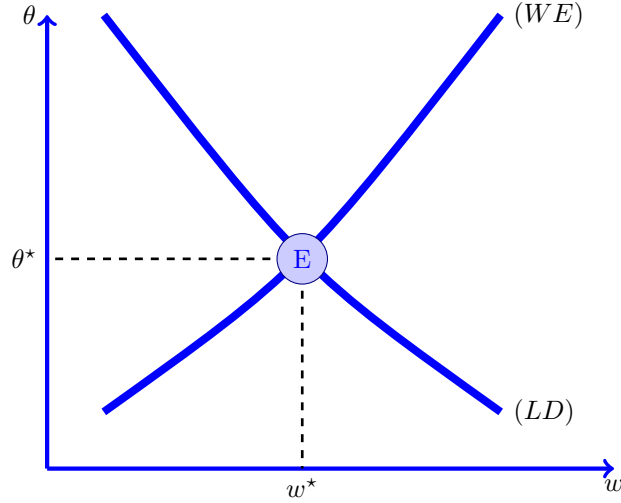
Plug into (1) and (2) to get

$$V_e - V_u = \frac{w - b}{r + p(\theta) + \delta}$$

and $\Pi_e = \frac{c}{q(\theta)}$ so that

$$w = b + c \frac{(r + \delta)}{a} \theta^{\frac{1}{2}} + c\theta \quad (WE)$$

One can check that (WE) gives θ as an increasing function of w while (LD) a decreasing one.



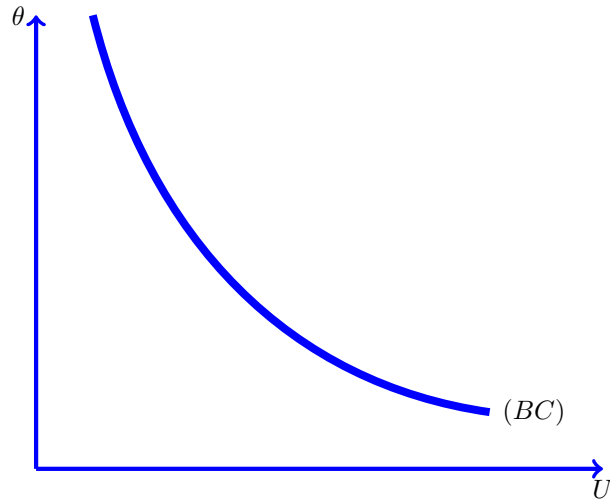
7 – Solve for the steady state unemployment where flows in and flows out from unemployment are equal. Use the formulas for $p(\theta)$ and $q(\theta)$ derived previously. This gives the Beveridge curve equation (BC) .

The law of motion of unemployment is $\dot{U} = \delta(1 - U) + p(\theta)U$ At the steady state, $\dot{U} = 0$ such that

$$U = \frac{\delta}{\delta - a\theta^{-\frac{1}{2}}}.$$

Hence the Beveridge curve

$$\theta = \left(\delta - \frac{\delta}{U} \right)^2 \frac{1}{a^2} \quad (BC)$$

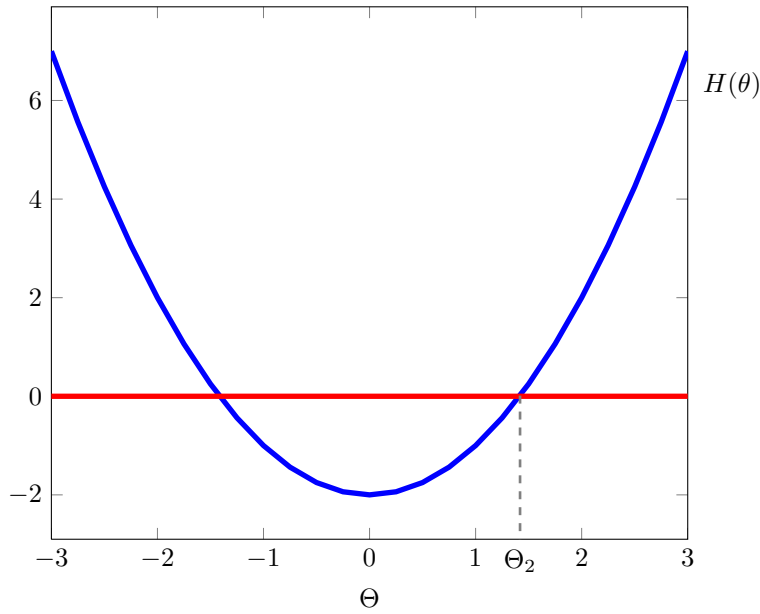


8 – By combining the equations (LD) and (WE) , one obtains an equation that relates θ to y , b , c , r , δ and a . Derive this equation.

That equation is $H(\theta) = c\theta + 2c\frac{(r+\delta)}{a}\theta^{\frac{1}{2}} - (y-b) = 0$.

Let's make the change of variable $\Theta = \sqrt{\theta}$. Then $H(\Theta)$ is a second order polynomial. H is convex that is convex as $c > 0$. $H(0) = -(y-b) < 0$ under the realistic assumption that $y > b$. Therefore, H has two roots and only the largest one Θ_2 is positive. $\theta = (\Theta_2)^2$ is the equilibrium level of θ . Note that an increase in a increases equilibrium θ .

Figure 6: Determination of equilibrium θ (with $\theta = (\Theta_2)^2$)



while an increase in b decreases it.

Once equilibrium θ is obtained, w and U are derived.

9 – Show how θ , U and w vary with b . Interpret the answers. Consider an increase in b . The two figures below shows how this shifts (WE), and therefore decreases θ (as already seen from Figure 7) and increases w and U .

10 – Show how θ , U and w vary with a . Interpret the answers. Consider an increase in a . The two figures below shows how this shifts (WE), LD and BC . The consequence is to increase θ (as already seen from Figure 8), increase w and decrease U .

Figure 7: Determination of equilibrium θ - an increase in b (with $\theta = (\Theta_2)^2$)

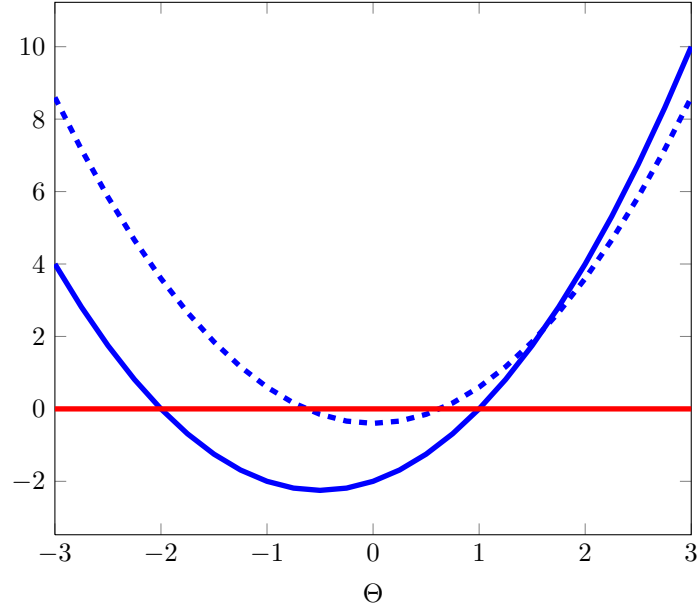


Figure 8: Determination of equilibrium θ - an increase in a (with $\theta = (\Theta_2)^2$)

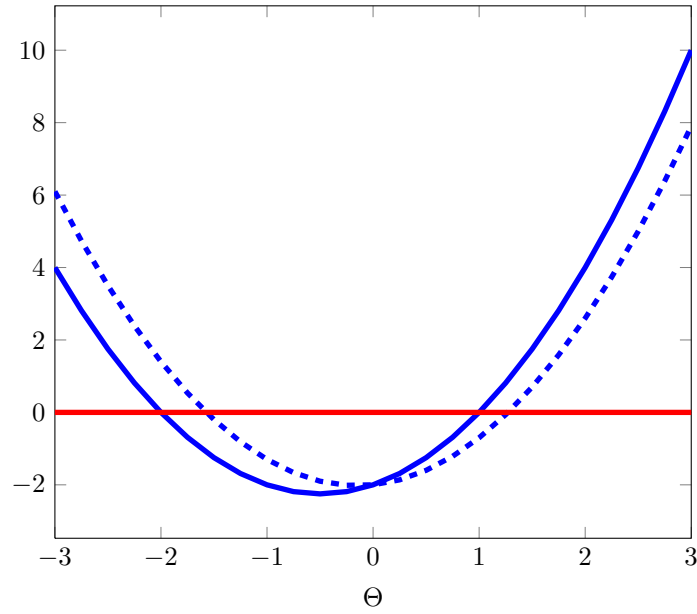


Figure 9: Impact of an increase in b

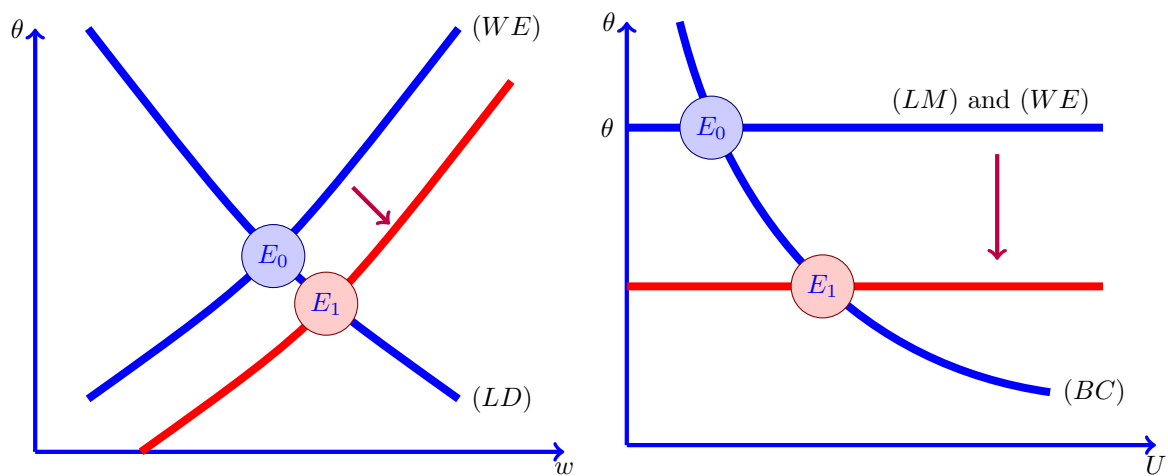


Figure 10: Impact of an increase in a

